

Target

Protein source

☐ RCSB PDB ID ☒ Local PDB file

Upload protein PDB

3PTB_clean.pdb
271.6KB

+

Ligand

Ligand SMILES

NC(=N)c1ccccc1

Pre-docked ligand SDF (recommended)

3ptb_C_BEN.sdf
1.5KB

+

☐ Automatically dock ligand with
AutoDock Vina

Docking box size (Å)

15.00

Vina exhaustiveness

8

Pull direction

Core chain

A

Core residue

45

-

+

Exit chain

A

Exit residue

84

-

+

SMD protocol

Pull distance (nm)

0.80

Pull velocity (nm/ps)

0.01



AMR-UNBIND v1.3.2

AMR ligand unbinding workflow: docking, explicit-solvent OpenMM SMD, and trajectory analysis

Docking Center Coordinates (Å)

⚠ These coordinates are specific to one PDB structure's own coordinate frame. The X/Y/Z values below are only an example (a hypothetical biotin pocket in PDB 1STP) — they are not a sensible default for any structure, and using them for one will very likely produce a nonsensical docking box (this is refused before it wastes a run, but it's better to avoid in the first place). Leave the override unchecked unless you know the correct coordinates for the specific structure you're running.

☐ Override auto-computed pocket center with these coordinates ?

Center X (example)

27.23

-

+

Center Y (example)

11.23

-

+

Center Z (example)

17.52

-

+

Run AMR-UNBIND

✓ Run complete

Current stage: complete — 0s in this stage — last update 19:19:26

Run directory: amr_workspace/runs/20260924T190808Z_3ptb_clean_2d185aef

Final displacement

0.192 nm

Peak |force|

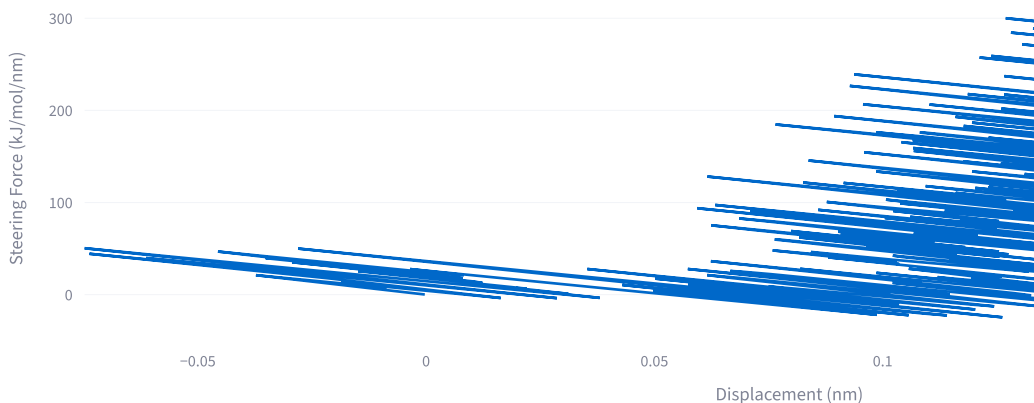
323.8 kJ/mol/...

Cumulative work

108.8 kJ/mol

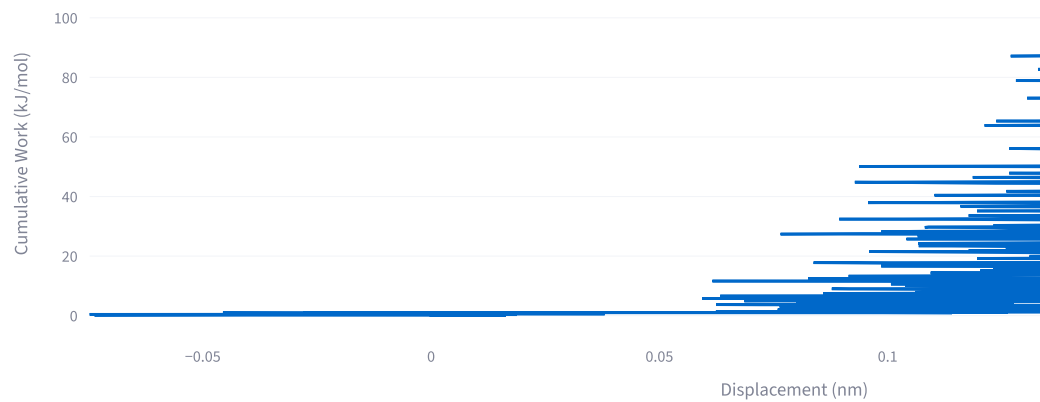
Force analysis

Steering Force vs. Ligand COM Displacement



Protocol work

Cumulative Protocol Work vs. Ligand COM Displacement



[Download analysis CSV](#)

[Download run metadata](#)

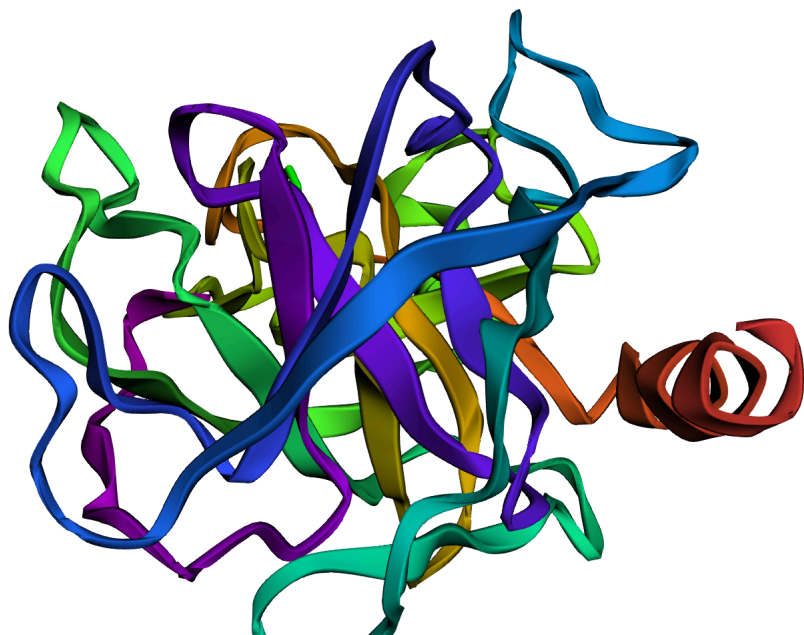
[Download ML-ready feature vector](#)

Prepared / final structure

REMARK	1 CREATED WITH OPENMM 8.6.1, 2026-09-24										
CRYST1	66.118	66.118	66.118	90.00	90.00	90.00	P 1			1	
ATOM	1	N	ILE	A	1	-7.984	9.806	20.422	1.00	0.00	N
ATOM	2	H	ILE	A	1	-7.017	9.860	20.709	1.00	0.00	H
ATOM	3	H2	ILE	A	1	-8.365	10.696	20.133	1.00	0.00	H
ATOM	4	H3	ILE	A	1	-8.456	9.401	21.218	1.00	0.00	H
ATOM	5	CA	ILE	A	1	-8.018	8.945	19.233	1.00	0.00	C
ATOM	6	HA	ILE	A	1	-7.166	9.122	18.576	1.00	0.00	H
ATOM	7	C	ILE	A	1	-9.266	9.257	18.506	1.00	0.00	C
ATOM	8	O	ILE	A	1	-10.372	9.067	19.004	1.00	0.00	O
ATOM	9	CB	ILE	A	1	-8.007	7.432	19.594	1.00	0.00	C
ATOM	10	HB	ILE	A	1	-8.877	7.202	20.209	1.00	0.00	H
ATOM	11	CG1	ILE	A	1	-6.766	6.989	20.438	1.00	0.00	C
ATOM	12	HG12	ILE	A	1	-6.832	5.934	20.704	1.00	0.00	H
ATOM	13	HG13	ILE	A	1	-6.796	7.644	21.309	1.00	0.00	H
ATOM	14	CG2	ILE	A	1	-8.175	6.640	18.260	1.00	0.00	C
ATOM	15	HG21	ILE	A	1	-8.113	5.604	18.592	1.00	0.00	H
ATOM	16	HG22	ILE	A	1	-7.273	6.840	17.681	1.00	0.00	H
ATOM	17	HG23	ILE	A	1	-9.107	6.907	17.762	1.00	0.00	H
ATOM	18	CD1	ILE	A	1	-5.419	7.155	19.723	1.00	0.00	C
ATOM	19	HD11	ILE	A	1	-5.375	6.369	18.968	1.00	0.00	H
ATOM	20	HD12	ILE	A	1	-5.648	8.014	19.092	1.00	0.00	H
ATOM	21	HD13	ILE	A	1	-4.528	7.229	20.347	1.00	0.00	H
ATOM	22	N	VAL	A	2	-9.129	9.761	17.265	1.00	0.00	N
ATOM	23	H	VAL	A	2	-8.184	9.748	16.909	1.00	0.00	H
ATOM	24	CA	VAL	A	2	-10.174	9.862	16.209	1.00	0.00	C
ATOM	25	HA	VAL	A	2	-11.093	9.968	16.785	1.00	0.00	H
ATOM	26	C	VAL	A	2	-10.463	8.572	15.558	1.00	0.00	C
ATOM	27	O	VAL	A	2	-9.512	7.966	15.149	1.00	0.00	O
ATOM	28	CB	VAL	A	2	-10.101	10.947	15.184	1.00	0.00	C
ATOM	29	HB	VAL	A	2	-9.268	10.741	14.512	1.00	0.00	H
ATOM	30	CG1	VAL	A	2	-11.478	11.128	14.607	1.00	0.00	C
ATOM	31	HG11	VAL	A	2	-11.379	11.959	13.908	1.00	0.00	H
ATOM	32	HG12	VAL	A	2	-12.314	11.314	15.281	1.00	0.00	H
ATOM	33	HG13	VAL	A	2	-11.695	10.310	13.921	1.00	0.00	H
ATOM	34	CG2	VAL	A	2	-9.767	12.269	15.982	1.00	0.00	C
ATOM	35	HG21	VAL	A	2	-10.455	12.336	16.825	1.00	0.00	H
ATOM	36	HG22	VAL	A	2	-8.736	12.273	16.335	1.00	0.00	H
ATOM	37	HG23	VAL	A	2	-9.897	13.170	15.382	1.00	0.00	H
ATOM	38	N	GLY	A	3	-11.700	8.073	15.543	1.00	0.00	N
ATOM	39	H	GLY	A	3	-12.488	8.595	15.900	1.00	0.00	H
ATOM	40	CA	GLY	A	3	-12.044	6.860	14.720	1.00	0.00	C
ATOM	41	HA2	GLY	A	3	-13.102	6.897	14.458	1.00	0.00	H
ATOM	42	HA3	GLY	A	3	-11.384	6.947	13.857	1.00	0.00	H
ATOM	43	C	GLY	A	3	-11.734	5.470	15.328	1.00	0.00	C
ATOM	44	O	GLY	A	3	-12.011	4.461	14.704	1.00	0.00	O
ATOM	45	N	GLY	A	4	-11.209	5.366	16.557	1.00	0.00	N
ATOM	46	H	GLY	A	4	-11.078	6.253	17.021	1.00	0.00	H
ATOM	47	CA	GLY	A	4	-11.013	4.251	17.380	1.00	0.00	C
ATOM	48	HA2	GLY	A	4	-10.427	4.472	18.273	1.00	0.00	H
ATOM	49	HA3	GLY	A	4	-10.426	3.522	16.821	1.00	0.00	H
ATOM	50	C	GLY	A	4	-12.273	3.571	17.888	1.00	0.00	C
ATOM	51	O	GLY	A	4	-13.405	3.873	17.532	1.00	0.00	O
ATOM	52	N	TYR	A	5	-12.004	2.638	18.822	1.00	0.00	N
ATOM	53	H	TYR	A	5	-11.039	2.528	19.098	1.00	0.00	H
ATOM	54	CA	TYR	A	5	-12.960	1.738	19.592	1.00	0.00	C
ATOM	55	HA	TYR	A	5	-13.995	2.071	19.509	1.00	0.00	H
ATOM	56	C	TYR	A	5	-12.639	1.691	21.065	1.00	0.00	C
ATOM	57	O	TYR	A	5	-11.521	1.842	21.471	1.00	0.00	O
ATOM	58	CB	TYR	A	5	-12.961	0.320	18.914	1.00	0.00	C
ATOM	59	HB2	TYR	A	5	-13.672	-0.321	19.435	1.00	0.00	H
ATOM	60	HB3	TYR	A	5	-13.394	0.473	17.925	1.00	0.00	H
ATOM	61	CG	TYR	A	5	-11.627	-0.359	18.847	1.00	0.00	C
ATOM	62	CD1	TYR	A	5	-11.174	-1.154	19.906	1.00	0.00	C
ATOM	63	HD1	TYR	A	5	-11.664	-1.273	20.861	1.00	0.00	H

ATOM	64	CD2	TYR	A	5	-10.908	-0.274	17.713	1.00	0.00	C
ATOM	65	HD2	TYR	A	5	-11.140	0.384	16.889	1.00	0.00	H
ATOM	66	CE1	TYR	A	5	-9.954	-1.836	19.852	1.00	0.00	C
ATOM	67	HE1	TYR	A	5	-9.547	-2.491	20.608	1.00	0.00	H
ATOM	68	CE2	TYR	A	5	-9.641	-0.900	17.633	1.00	0.00	C
ATOM	69	HE2	TYR	A	5	-9.043	-0.717	16.753	1.00	0.00	H
ATOM	70	CZ	TYR	A	5	-9.184	-1.771	18.672	1.00	0.00	C
ATOM	71	OH	TYR	A	5	-8.009	-2.338	18.529	1.00	0.00	O
ATOM	72	HH	TYR	A	5	-7.989	-3.177	18.995	1.00	0.00	H
ATOM	73	N	THR	A	6	-13.667	1.653	21.925	1.00	0.00	N
ATOM	74	H	THR	A	6	-14.617	1.672	21.582	1.00	0.00	H
ATOM	75	CA	THR	A	6	-13.356	1.433	23.376	1.00	0.00	C
ATOM	76	HA	THR	A	6	-12.825	2.327	23.704	1.00	0.00	H
ATOM	77	C	THR	A	6	-12.548	0.150	23.579	1.00	0.00	C
ATOM	78	O	THR	A	6	-13.059	-0.947	23.275	1.00	0.00	O
ATOM	79	CB	THR	A	6	-14.658	1.360	24.304	1.00	0.00	C
ATOM	80	HB	THR	A	6	-15.436	0.736	23.863	1.00	0.00	H
ATOM	81	OG1	THR	A	6	-15.263	2.590	24.111	1.00	0.00	O
ATOM	82	HG1	THR	A	6	-15.512	2.707	23.191	1.00	0.00	H
ATOM	83	CG2	THR	A	6	-14.541	1.192	25.841	1.00	0.00	C
ATOM	84	HG21	THR	A	6	-15.533	1.255	26.287	1.00	0.00	H
ATOM	85	HG22	THR	A	6	-14.085	0.236	26.099	1.00	0.00	H
ATOM	86	HG23	THR	A	6	-13.826	1.942	26.179	1.00	0.00	H
ATOM	87	N	CYS	A	7	-11.319	0.340	24.146	1.00	0.00	N
ATOM	88	H	CYS	A	7	-11.154	1.296	24.429	1.00	0.00	H
ATOM	89	CA	CYS	A	7	-10.366	-0.717	24.410	1.00	0.00	C
ATOM	90	HA	CYS	A	7	-10.042	-1.133	23.456	1.00	0.00	H
ATOM	91	C	CYS	A	7	-10.890	-1.908	25.247	1.00	0.00	C
ATOM	92	O	CYS	A	7	-10.689	-3.107	24.906	1.00	0.00	O
ATOM	93	CB	CYS	A	7	-9.139	-0.170	25.028	1.00	0.00	C
ATOM	94	HB2	CYS	A	7	-8.429	-0.979	25.201	1.00	0.00	H
ATOM	95	HB3	CYS	A	7	-9.446	0.094	26.039	1.00	0.00	H
ATOM	96	SG	CYS	A	7	-8.499	1.319	24.283	1.00	0.00	S
ATOM	97	N	GLY	A	8	-11.555	-1.518	26.341	1.00	0.00	N
ATOM	98	H	GLY	A	8	-11.703	-0.531	26.491	1.00	0.00	H
ATOM	99	CA	GLY	A	8	-12.076	-2.389	27.408	1.00	0.00	C
ATOM	100	HA2	GLY	A	8	-12.375	-3.390	27.096	1.00	0.00	H
ATOM	101	HA3	GLY	A	8	-12.976	-1.980	27.866	1.00	0.00	H
ATOM	102	C	GLY	A	8	-10.944	-2.447	28.440	1.00	0.00	C
ATOM	103	O	GLY	A	8	-9.808	-2.677	28.023	1.00	0.00	O
ATOM	104	N	ALA	A	9	-11.137	-2.234	29.738	1.00	0.00	N
ATOM	105	H	ALA	A	9	-12.049	-1.913	30.029	1.00	0.00	H
ATOM	106	CA	ALA	A	9	-10.055	-2.278	30.739	1.00	0.00	C
ATOM	107	HA	ALA	A	9	-9.295	-1.542	30.480	1.00	0.00	H
ATOM	108	C	ALA	A	9	-9.278	-3.607	30.773	1.00	0.00	C
ATOM	109	O	ALA	A	9	-9.946	-4.634	30.496	1.00	0.00	O
ATOM	110	CB	ALA	A	9	-10.725	-2.073	32.135	1.00	0.00	C
ATOM	111	HB1	ALA	A	9	-11.433	-2.883	32.310	1.00	0.00	H
ATOM	112	HB2	ALA	A	9	-9.974	-2.156	32.920	1.00	0.00	H
ATOM	113	HB3	ALA	A	9	-11.224	-1.116	32.289	1.00	0.00	H
ATOM	114	N	ASN	A	10	-7.936	-3.555	31.150	1.00	0.00	N
ATOM	115	H	ASN	A	10	-7.579	-2.624	31.308	1.00	0.00	H
ATOM	116	CA	ASN	A	10	-7.026	-4.705	31.265	1.00	0.00	C
ATOM	117	HA	ASN	A	10	-6.179	-4.202	31.732	1.00	0.00	H
ATOM	118	C	ASN	A	10	-6.657	-5.239	29.899	1.00	0.00	C
ATOM	119	O	ASN	A	10	-5.879	-6.206	29.871	1.00	0.00	O
ATOM	120	CB	ASN	A	10	-7.636	-5.746	32.257	1.00	0.00	C
ATOM	121	HB2	ASN	A	10	-6.874	-6.525	32.289	1.00	0.00	H
ATOM	122	HB3	ASN	A	10	-8.397	-6.309	31.716	1.00	0.00	H
ATOM	123	CG	ASN	A	10	-8.084	-5.254	33.629	1.00	0.00	C
ATOM	124	OD1	ASN	A	10	-7.279	-5.204	34.517	1.00	0.00	O
ATOM	125	ND2	ASN	A	10	-9.279	-4.756	33.766	1.00	0.00	N
ATOM	126	HD21	ASN	A	10	-9.461	-4.292	34.645	1.00	0.00	H
ATOM	127	HD22	ASN	A	10	-9.957	-4.635	33.027	1.00	0.00	H
ATOM	128	N	THR	A	11	-7.164	-4.707	28.791	1.00	0.00	N
ATOM	129	H	THR	A	11	-7.759	-3.896	28.880	1.00	0.00	H

ATOM	130	CA	THR	A	11	-6.790	-5.179	27.412	1.00	0.00	C
ATOM	131	HA	THR	A	11	-6.711	-6.266	27.443	1.00	0.00	H
ATOM	132	C	THR	A	11	-5.397	-4.751	26.903	1.00	0.00	C
ATOM	133	O	THR	A	11	-4.943	-5.285	25.890	1.00	0.00	O
ATOM	134	CB	THR	A	11	-7.838	-4.883	26.275	1.00	0.00	C
ATOM	135	HB	THR	A	11	-7.394	-5.195	25.330	1.00	0.00	H
ATOM	136	OG1	THR	A	11	-8.031	-3.529	26.125	1.00	0.00	O
ATOM	137	HG1	THR	A	11	-8.710	-3.309	26.767	1.00	0.00	H
ATOM	138	CG2	THR	A	11	-9.137	-5.708	26.388	1.00	0.00	C
ATOM	139	HG21	THR	A	11	-9.689	-5.484	27.301	1.00	0.00	H
ATOM	140	HG22	THR	A	11	-8.801	-6.745	26.400	1.00	0.00	H
ATOM	141	HG23	THR	A	11	-9.737	-5.646	25.480	1.00	0.00	H
ATOM	142	N	VAL	A	12	-4.806	-3.819	27.629	1.00	0.00	N
ATOM	143	H	VAL	A	12	-5.354	-3.564	28.438	1.00	0.00	H
ATOM	144	CA	VAL	A	12	-3.602	-3.108	27.210	1.00	0.00	C
ATOM	145	HA	VAL	A	12	-3.098	-3.770	26.506	1.00	0.00	H
ATOM	146	C	VAL	A	12	-2.851	-2.900	28.513	1.00	0.00	C
ATOM	147	O	VAL	A	12	-2.777	-1.835	29.13			



Optional ChimeraX rendering

ChimeraX REST port

3000

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Ask ChimeraX to render movie